# Weekly Record 7: Technology and Data

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Chapter 13, Using Technology to Manage Information · Company B: Bayan Sulu · Reduced record, Fields 1–2 only (Fields 3–5 absorbed into A3 Photo Essay) · Develops into Capstone Section 6, Governance, verification and assurance.

## Field 1 · Concept

Key terms from Chapter 13 [1]:

- Data and information: data are raw facts, like one sale at a till. Information is data that has been organised so a manager can use it, for example "waffle sales in Almaty went down this month".
- Management information system (MIS): the methods and equipment that give managers information about every part of the business.
- Transaction processing system (TPS): handles the daily work like orders, invoices and deliveries. The other systems get their raw data from it.
- Batch and online processing: batch updates data in groups, for example once a night. Online (real-time) updates it as soon as it comes in.
- Decision support system (DSS): lets managers try "what if" models before they decide.
- Expert system: uses AI to give advice, a bit like asking a human expert.
- Data warehouse and data mart: a warehouse puts all the company's databases in one place. A data mart is a smaller part of it for one area, like marketing.
- Data mining: looking through big data sets for hidden patterns that help predict what people will do.
- Knowledge management: collecting and sharing what the company knows, so it doesn't disappear when people leave.
- Data security: protecting data from viruses, hackers, theft and simple mistakes, with things like firewalls, encryption, passwords, backups and staff training.

Concept Check: Why are privacy advocates worried about data warehouses and data mining?

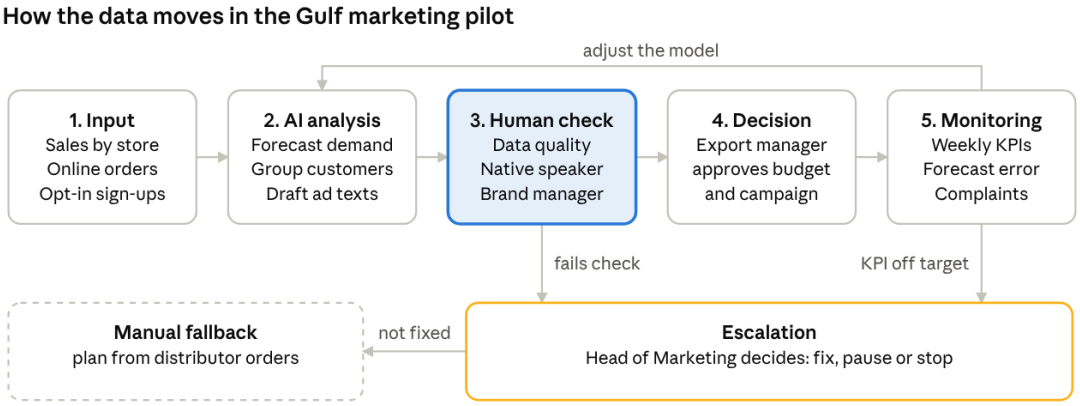
When a company joins a lot of databases together, it can find out much more about one person than that person ever agreed to share. Then data mining can guess things about them, like their income or health, and the company might treat them differently because of it. Most customers have no idea this is happening. In my opinion, the more a company can do with data, the more rules it needs about what it is allowed to do.

## Field 2 · Application

For the practice task I made an AI/data workflow and a risk register for Bayan Sulu's marketing pilot in the Gulf.

It continues my Week 6 record. There I suggested a small AI marketing test for halal-seeking families in the Gulf. This week I wanted to show how the data actually moves, where people check it, and what

happens if something goes wrong. Using the chapter's words, the TPS collects the sales data, a marketing data mart keeps it, and the AI works like a decision support system. It gives options, but managers make the decision.




Gulf marketing pilot · data workflow, 5 steps, escalation and fallback

The AI never acts by itself. At step 3 a person can stop anything it makes, and at step 5 problems go up to escalation or to the manual plan.

Risk register. I gave every risk one owner, a trigger for when to escalate, and a manual backup. The numbers are my own suggestions for the pilot, not real company data.

| Risk | What could go wrong | How we control it | Owner | Escalation trigger and backup |
|---|---|---|---|---|
| Privacy | Customer data is used without consent, or data from public posts is reused for ads | Only use loyalty sign-ups where people agreed; don't take data from social media [2] | Data protection officer | Any complaint or missing consent: stop using that data and review it within 48 hours |
| Security | The customer list gets leaked or hacked | Encryption, only named staff have access, two-step login, monthly backup [1], [3] | IT manager | Any breach: report it under the 2026 breach register rules [3] and switch to an offline list |
| Bias | AI shows offers only to richer areas or to one language group | Check every month that all groups and languages get a fair share | Marketing manager | A group gets less than half its share: fix it by hand |

| Risk | What could go wrong | How we control it | Owner | Escalation trigger and backup |
|---|---|---|---|---|
| Reliability | A wrong forecast, a bad translation, or an ad with a false claim (for example about health) | Compare the forecast with real orders; a native speaker and the brand manager approve every ad | Export manager | Forecast error over 25% for 2 months: go back to planning from distributor orders |
| Accountability | Nobody knows who approved a campaign or a price | Every approval is saved with a name and date; AI never publishes ads or sets prices by itself [2], [4] | Head of Marketing | A decision with no name on it: pause the campaign until someone signs off |

I built the register around two rules. Kazakhstan's 2026 law does not allow decisions made only by automated processing [2], and the NIST framework says every AI system needs clear human roles [4]. So in this workflow the AI can only suggest. A named person has to act.

**Sources**


[1] L. J. Gitman et al., *Introduction to Business*, 2nd ed. Houston, TX, USA: OpenStax, 2026, ch. 13. [Online]. Available: https://openstax.org/details/books/introduction-business-2e

[2] A. Shyngyssov and B. Kadyrov, "Republic of Kazakhstan Data Protection Law: Amendments in 2026," Morgan Lewis LawFlash, Jul. 14, 2026. [Online]. Available: https://www.morganlewis.com/pubs/2026/07/republic-of-kazakhstan-data-protection-law-amendments-in-2026

[3] Baker McKenzie, "Kazakhstan: New Amendments to Personal Data Legislation," Jul. 21, 2026. [Online]. Available: https://www.bakermckenzie.com/en/insight/publications/2026/07/kazakhstan-new-amendments-to-personal-data-legislation

[4] National Institute of Standards and Technology, "Artificial Intelligence Risk Management Framework (AI RMF 1.0)," NIST AI 100-1, Jan. 2023, doi: 10.6028/NIST.AI.100-1.